

3.6 Interpolation

When three tabular values are available, the interpolating polynomial takes its simplest form. The correction to the tabular argument is obtained from the second difference of the function values.

$$\Delta n_0 = - \frac{2y_2 + n_0(a + b + cn_0)}{a + b + 2cn_0}$$

where a and b denote the first differences and c the second difference. The iteration converges rapidly whenever the tabular interval is small compared with the curvature of the function.

$$y = y_3 + \frac{n}{2}(B + C) + \frac{n^2}{2}F$$

which may also be written in terms of the accumulated differences.